

Computational Microelectronics

Lecture 6

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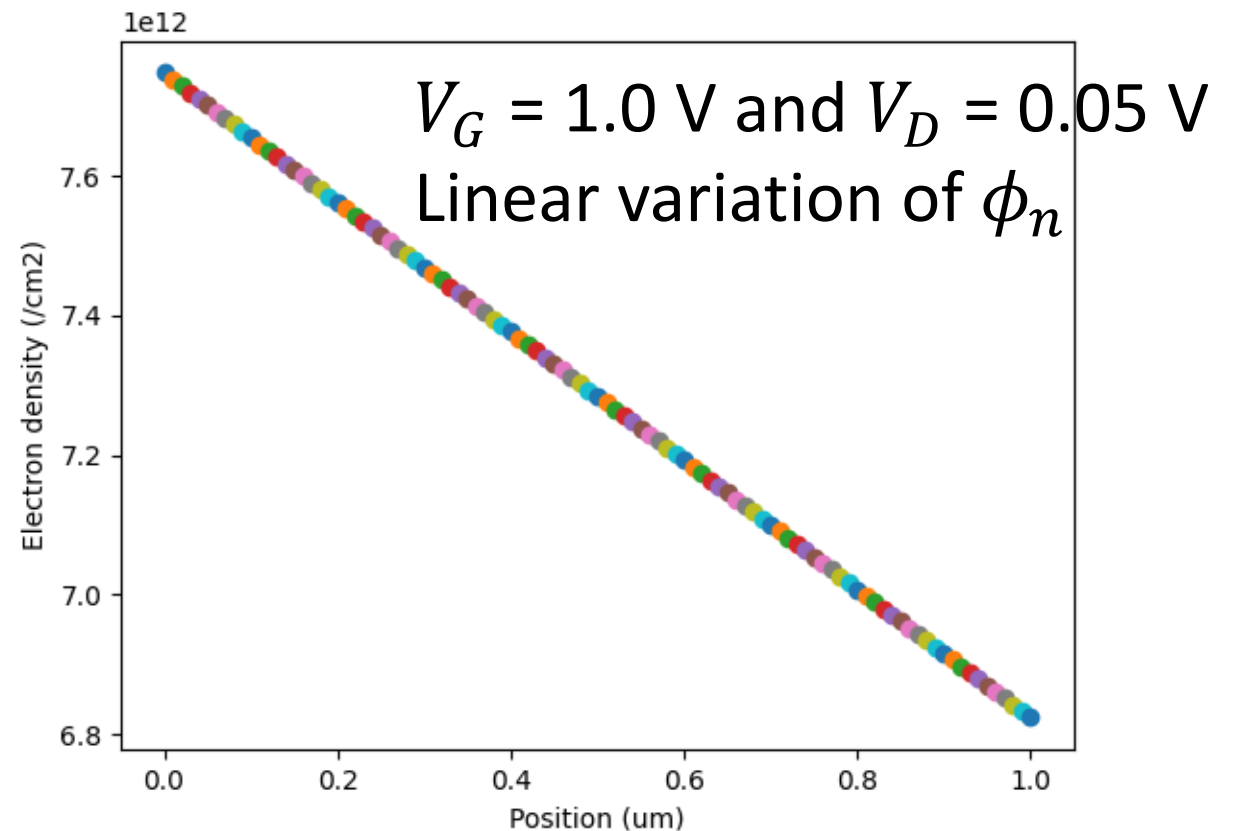
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L6

Check the electron continuity.

- Even when $V_D = 0.05$ V, the electron density is not uniform.
 - From the continuity, $\frac{d}{dx}J_{n,x} = \frac{d}{dx}(Q_n v_{n,x}) = 0$, we need a uniform value of $Q_n v_{n,x}$.
 - Recall that $\frac{d\phi_n}{dx} = \frac{V_D - V_S}{L_G}$.
($v_{n,x}$ is $\mu_n \frac{V_D - V_S}{L_G}$.)
 - But Q_n is not uniform.
- Linear variation of ϕ_n
 - It fails.

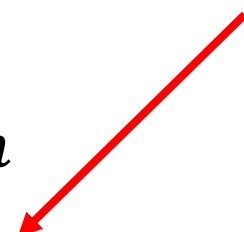


$\phi_n \Rightarrow Q_n$, now, $Q_n \Rightarrow \phi_n$

- MOS equation, $\phi_n \Rightarrow Q_n$
- Electron continuity equation, $Q_n \Rightarrow \phi_n$

$$Q_n v_{n,x} = Q_n \mu_n \frac{d\phi_n}{dx} = J_{n,x}$$

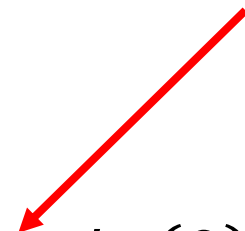
A constant



- When Q_n is known, ϕ_n satisfies

$$\frac{d\phi_n}{dx} = \frac{J_{n,x}}{Q_n \mu_n}$$

Constant mobility



- Integrating from 0 to x ,

$$\phi_n(x) = \phi_n(0) + \int_0^x \frac{J_{n,x}}{Q_n(x') \mu_n(x')} dx' = \phi_n(0) + \frac{J_{n,x}}{\mu_n} \int_0^x \frac{1}{Q_n(x')} dx'$$

(We don't know $J_{n,x} \dots$)

Integration from 0 to L_G

- Using the boundary condition,
 - We can find an expression for $J_{n,x}$:

$$V_D = V_S + \frac{J_{n,x}}{\mu_n} \int_0^{L_G} \frac{1}{Q_n(x')} dx'$$

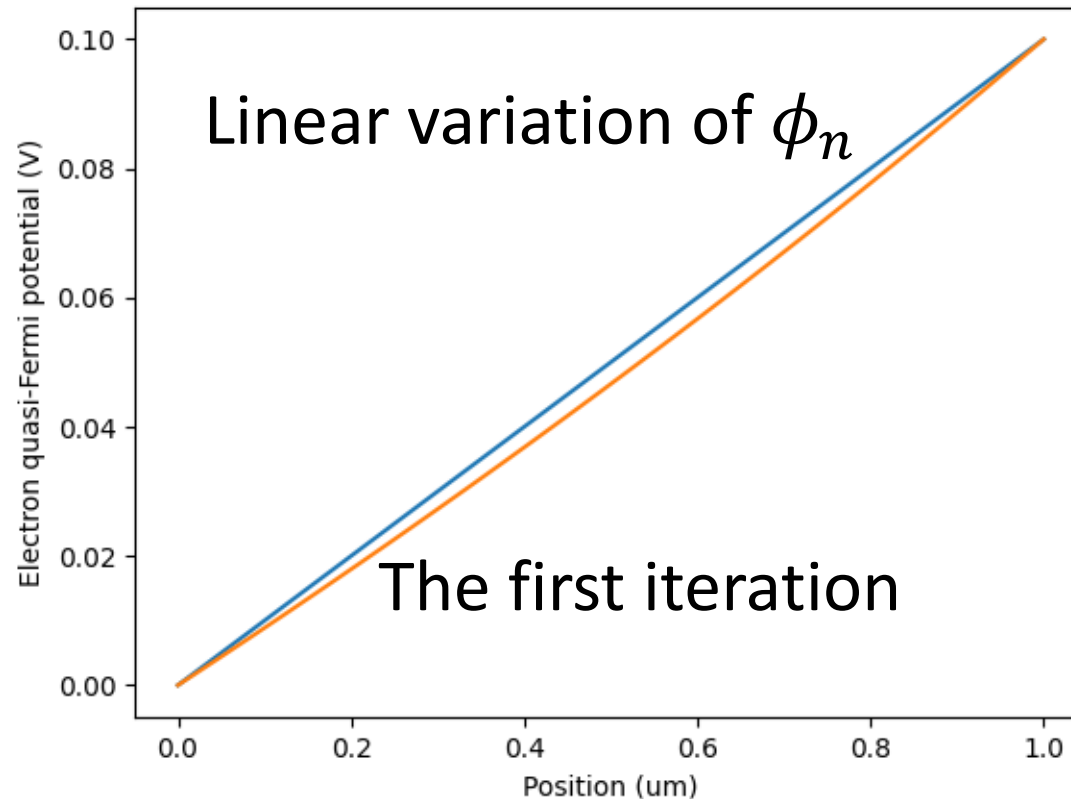
- Therefore,

$$\phi_n(x) = V_S + \frac{V_{DS}}{\int_0^{L_G} \frac{1}{Q_n(x')} dx'} \int_0^x \frac{1}{Q_n(x')} dx'$$

(Numerical integrations are needed. At each segment, Q_n at the left point is taken.)

The first iteration

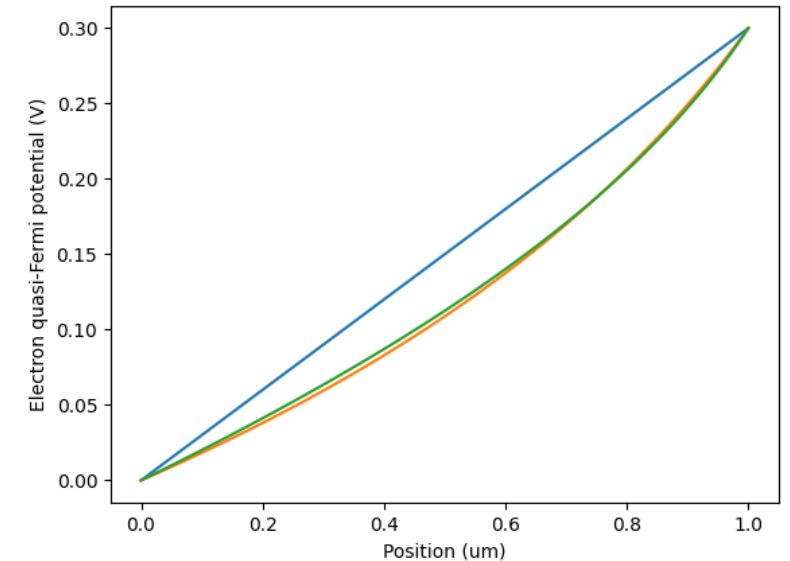
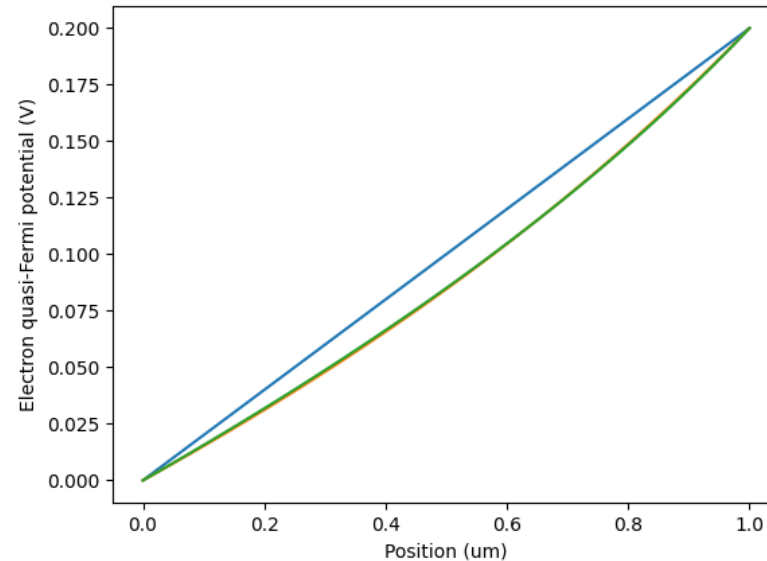
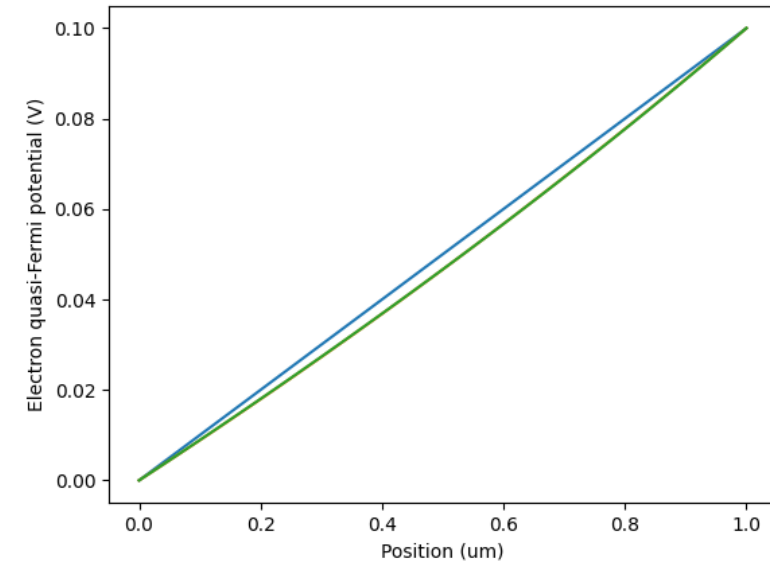
- Recall that we originally assumed $\phi_n(x) = V_S + \frac{V_{DS}}{L_G} x$.
 - Now, the solution of the electron continuity equation deviates from the assumed profile.



$$V_G = 1.0 \text{ V and } V_D = 0.1 \text{ V}$$

Repeated iterations

- Three V_D values, 0.1 V, 0.2 V, and 0.3 V
 - Orange: The first iteration
 - Green: The second iteration



Calculated drain current

- The drain current is $-J_{n,x}W$. (W is the channel width.)
 - When μ_n is fixed, it can be immediately calculated.

$$I_D = -J_{n,x}W = -\frac{\mu_n V_{DS} W}{\int_0^{L_G} \frac{1}{Q_n(x')} dx'}$$

- Convergence of I_D after iterations (This is just an example.)

6.260461008022095e-06

6.260568954173386e-06

6.260568946123915e-06

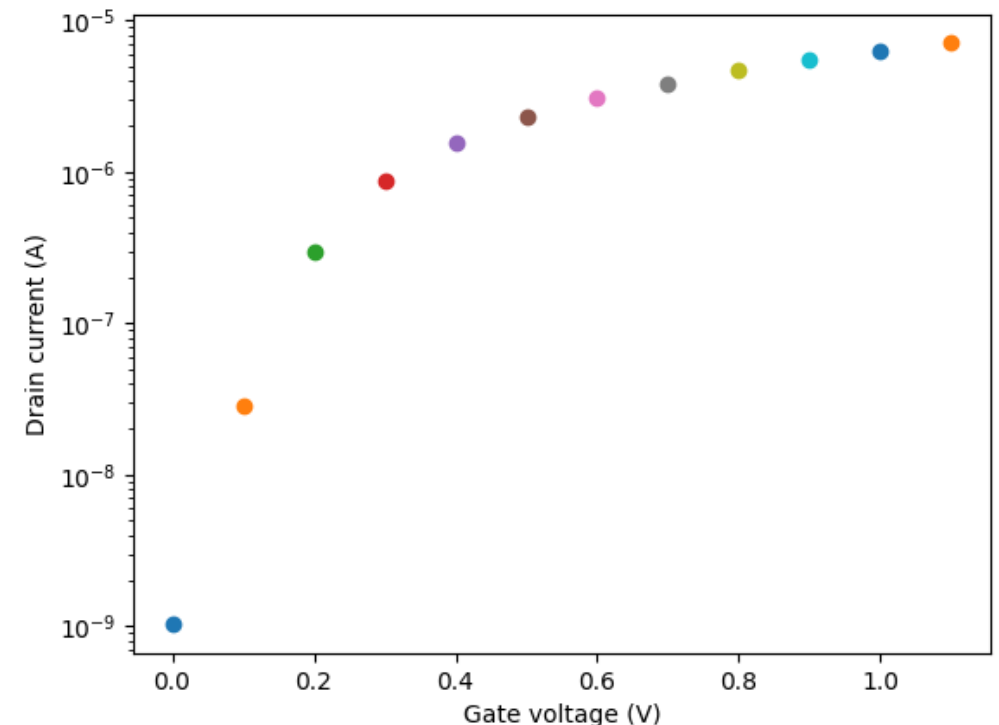
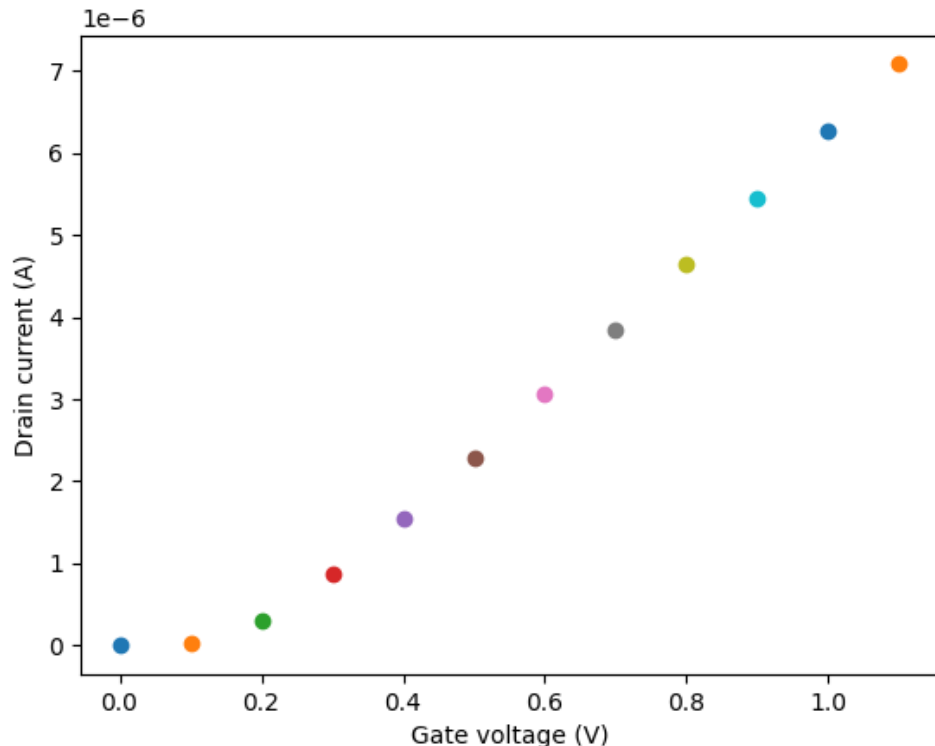
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Check your results.

- Φ_m is 4.03 eV. N_a is $1.0 \times 10^{18} \text{ cm}^{-3}$. t_{ox} is 1.2 nm. μ_n is 300 $\text{cm}^2 \text{ V}^{-1} \text{ s}^{-1}$. $L_G = W = 1 \text{ } \mu\text{m}$.
 - IV curve at $V_D = 0.01 \text{ V}$(Numbers are dependent on material parameters.)



Thank you!